

The Flower

Floral Whorls

A typical flower has four whorls on the thalamus: **calyx** (sepals), **corolla** (petals) — accessory; **androecium** (stamens) and **gynoecium** (carpels) — reproductive. Undifferentiated calyx+corolla = **perianth**.

Symmetry

Actinomorphic (radial, Mustard), **Zygomorphic** (bilateral, Pea), **Asymmetric** (Canna).

Position on Thalamus

Hypogynous — superior ovary (Mustard); **Perigynous** — half-inferior ovary (Plum); **Epigynous** — inferior ovary (Guava).

Aestivation

Valvate (Calotropis), **Twisted** (China rose), **Imbricate** (Gulmohur), **Vexillary/Papilionaceous** (Pea).

Androecium & Gynoecium

Stamens: **monadelphous**, **diadelphous**, **polyadelphous**, or **epipetalous**. Carpels: **apocarpous** (free, Lotus) or **syncarpous** (fused, Mustard).

Placentation

Marginal (Pea), **Axile** (Tomato), **Parietal** (Mustard), **Free central** (Dianthus), **Basal** (Sunflower).

Structure of the Androecium in Detail

Each stamen consists of a slender **filament** and a terminal **anther**, typically bilobed, with each lobe having two chambers (theca) producing pollen grains. The point of attachment between filament and anther is the connective.

Structure of the Gynoecium in Detail

Each carpel comprises a swollen basal **ovary** containing ovules on the placenta, an elongated **style**, and a terminal **stigma** that receives pollen grains during pollination.

Floral Diagrams and Formula

A **floral diagram** is a schematic cross-sectional view showing the arrangement of floral parts, while a **floral formula** uses symbols (K, C, A, G) with numbers to represent each whorl symbolically — an essential taxonomic shorthand.

Types of Placentation

Type	Description	Example
Marginal	Ridge along ventral suture	Pea
Axile	Central axis, multilocular ovary	Tomato, China rose
Parietal	Peripheral wall, unilocular	Mustard, Argemone
Free Central	Central axis, no septa	Dianthus
Basal	Base of ovary, single ovule	Sunflower

5 Types of Placentation

- Marginal
- Axile
- Parietal
- Free Central

- Basal